Zer0pa / ZPE-Ink

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ZPE-Ink V0.0: DETERMINISTIC DIGITAL INK CODEC: .zpink Streams | Stroke Compression | 5x+ vs Float32 | Rust | WASM | Swift | C# Cross-Runtime

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README | Code of conduct | Contributing | License | Security

ZPE-Ink

Deterministic digital-ink codec centered on the `.zpink` packet format. This repo is a private staging snapshot with a current proof subset and rerun surface. It is not release-ready.

Status in plain language:

- What this is: a staged `.zpink` codec with a Python install surface and repo-local bindings.
- Proven: structured-tier compression exceeds 5x vs raw float32 and cross-runtime parity logs pass locally.
- Blocked: sovereign release surface is `FAIL` and blind-clone verification is `INCONCLUSIVE`.

Sovereign release surface: `proofs/reruns/contradiction_resolution_local/contradiction_resolution_manifest.json`.

Prereqs for local verification: Python 3.11+, Rust toolchain, and `wasm32-unknown-unknown` target for binding checks.

WHAT THIS IS

ZPE-Ink is the staged codec surface for `.zpink` stream encoding and decoding. The installable release unit is the Python package under `code/`. The Rust/WASM/Swift/C# bindings are repo-local source surfaces and are not part of the pip install unit.

QUICKSTART AND LICENSE

```
python -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -e './code[dev]'
python -m pytest code/tests -q
python -m zpe_ink demo
python -m zpe_ink verify-roundtrip
```

Authoritative artifact: `proofs/reruns/benchmark_freeze_local/claim_scope_map.json`

Key coordinates	Value
Repository URL	`https://github.com/Zer0pa/ZPE-Ink`
Contact	`architects@zer0pa.ai`
Current verdict	`INCONCLUSIVE` (release surface `FAIL`)
Claim family	`structured-tier-only` (>5x vs raw float32

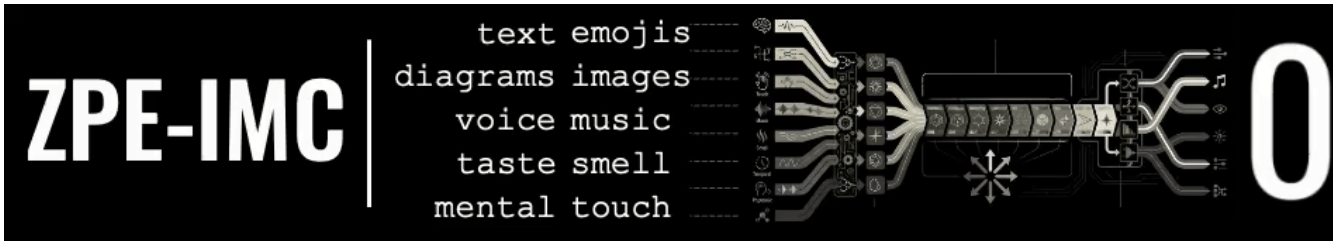
How to read this table: it reports the latest measured ratios and boundaries; it does not imply release readiness.

Authority snapshot	Value	Evidence
Structured tier ratio	`5.590209480060199x`	`proofs/reruns/benchmark_freeze_local/baseline_results.json`
Structured tier best comparator	`brotli 6.825565026256283x`	`proofs/reruns/benchmark_freeze_local/baseline_results.json`
Hard corpus ratios	`MathWriting 1.0944x`, `CROHME 1.3015x`	`proofs/reruns/benchmark_freeze_local/baseline_results.json`

Authority snapshot	Value	Evidence
Release surface	`FAIL` (handoff `NO-GO`)	`proofs/reruns/contradiction_resolution_local/contradiction_resolution_manifest.json`
Blind clone	`INCONCLUSIVE` (npm probe failure)	`proofs/reruns/phase3_external/blind_clone_verdict.json`
Non-Latin corpus	`Calliar 2.7746x`	`proofs/reruns/phase3_external/calliar_benchmark.json`

How to read this table: these are the current authority anchors; any conflict keeps the repo `INCONCLUSIVE`.

Proof anchor	Purpose	Current truth
`proofs/reruns/contradiction_resolution_local/contradiction_resolution_manifest.json`	Sovereign release surface	`release_surface_ve`
`proofs/reruns/benchmark_freeze_local/claim_scope_map.json`	Claim boundary	`structured-tier-on`
`proofs/reruns/benchmark_freeze_local/baseline_results.json`	Structured -tier ratios	`appendix_all_pass=`
`proofs/logs/20260321_technical_alignment_cross_runtime.json`	Cross-runtime parity log	`status=pass`
`proofs/logs/20260321_technical_alignment_binding_contracts.json`	Binding contract check	`status=PASS`
`proofs/reruns/phase3_external/blind_clone_verdict.json`	Blind-clone gate	`verdict=INCONCLUSI`



MODALITY STATUS SNAPSHOT

Surface	Status	Evidence
Structured-tier transport	`PASS` (>5x vs raw float32)	`proofs/reruns/benchmark_freeze_local/claim_scope_map.json`
Hard-corpus transport	`FAIL` (below best comparators)	`proofs/reruns/benchmark_freeze_local/baseline_results.json`
Release surface	`FAIL` / `INCONCLUSIVE`	`proofs/INK_WAVE1_RELEASE_READINESS_REPORT.md`
Blind clone	`INCONCLUSIVE`	`proofs/reruns/phase3_external/blind_clone_verdict.json`
Cross-runtime parity (current)	`PASS`	`proofs/logs/20260321_technical_alignment_cross_runtime.json`
Contract alignment (repo-local)	`PASS`	`proofs/logs/20260321_technical_alignment_binding_contracts.json`

REPO SHAPE

Area	Purpose
`code/`	Installable Python package, tests, bindings, scripts
`docs/`	Architecture, contracts, support, legal boundaries, doc registry
`proofs/`	Reruns, logs, runbooks, current proof anchors
`executable/`	Local smoke and verification entry points

ZPE One Codec… …to rule them all | ZER0PA

OPEN RISKS (NON-BLOCKING)

Release blockers:

- Sovereign release surface remains `FAIL` / `INCONCLUSIVE` while the handoff manifest remains `NO-GO` .
- UNIPEN access remains unresolved; IAM remains registration-gated; cross-script authority is still restricted to Calliar-only evidence.
- Blind clone is still `INCONCLUSIVE` until the gate-a resource probe rerun is recorded on the updated code.

Constraints and technical debt:

- Primitive-token branch is candidate-only; Calliar bounded fidelity fails with large Hausdorff error.
- Telemetry reruns are incomplete in this shell without `COMET_API_KEY` and `RUNPOD_API_KEY` .

CONTRIBUTING, SECURITY, SUPPORT

Route	Target
Documentation index	`docs/README.md`
Auditor path	`AUDITOR_PLAYBOOK.md`
Governance rules	`GOVERNANCE.md`
Release gate rules	`RELEASING.md`

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

Zer0pa-Architect-Prime

Languages

Python 87.0% Rust 7.5% Swift 3.7% Shell 1.0% C# 0.3% Makefile 0.3% JavaScript 0.2%